

Beyond the Predicted Class: Calibrated Explanations for Real Multiclass Decisions*

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Editor: Ernst Ahlberg, Ulf Johansson, Henrik Boström, Alberto Carlevaro, Johan Hallberg Szabadváry and Lars Carlsson

Abstract

Calibrated Explanations provide uncertainty-aware feature importance and probabilistic predictions for binary classification, but existing multiclass extensions explain only the most probable class. This limitation reduces interpretability, especially when prediction confidence is low. We extend Calibrated Explanations to full multiclass support using a one-vs-rest strategy, generating calibrated probability intervals and class-specific feature importance for all classes. Experiments on four datasets show lower repeated-run variance than calibrated SHAP and LIME, while uncalibrated SHAP and LIME achieved lower variance in the robustness evaluation. A user study with 22 experts found that the proposed explanation format received the highest mean trust score and strongest overall preference ranking, while also influencing participants’ decisions.

Keywords: Explainable AI, Calibrated Explanations, Uncertainty Quantification, Multiclass Classification, Predictive Maintenance, Feature Importance, Model Calibration, Human-AI Trust

1. Introduction

Machine learning (ML) and deep learning (DL) models enable accurate predictions, which is essential in sensitive domains such as healthcare, marketing, education, finance, agriculture, and manufacturing (An et al., 2023; Soofi and Awan, 2017). Despite improvements in predictive accuracy and the widespread adoption of ML and DL models for failure prediction in manufacturing, these models often function as opaque “black boxes,” providing limited insight into their decision-making (DM) processes (Sinha et al., 2023). This lack of interpretability hinders trust and practical adoption, creating an imperative for models that generate human-understandable explanations accessible to both operators and domain experts (Sinha et al., 2023; Balakir et al., 2020). Explainable AI (XAI) addresses this requirement by illuminating how AI systems derive conclusions, predictions, or recommendations, thereby fostering trust and enabling effective human-AI collaboration (Ghadekar et al., 2024).

State-of-the-art XAI techniques include SHapley Additive exPlanations (SHAP) and Local Interpretable Model Agnostic Explanations (LIME). SHAP leverages game-theoretic Shapley values to fairly attribute feature contributions to model predictions (Lundberg and

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Lee, 2017), while LIME approximates black-box behavior using interpretable surrogates such as linear regression or decision trees (Ribeiro et al., 2016). Calibrated Explanations (CE) represents an emerging approach that enhances interpretability by aligning model output with reliable explanations of feature importance (FI) (Löfström et al., 2024).

A parallel challenge concerning probabilistic predictions is that while ML and DL models output class probabilities, they rarely provide associated confidence scores (Beattie et al., 2022). Overconfident yet incorrect predictions risk unnecessary maintenance interventions or, conversely, catastrophic failures when critical conditions are underestimated (Beattie et al., 2022). Post-hoc calibration techniques, including Platt scaling (Platt et al., 1999), isotonic regression (Zadrozny and Elkan, 2001), and Venn–Abers (VA) predictors (Vovk and Petej, 2012), can be applied to model output to generate well-calibrated probabilistic predictions.

Furthermore, while existing XAI methods provide FI explanations, they typically do not quantify the uncertainty associated with these feature contributions (Löfström et al., 2024). Uncertainty quantification (UQ) for explanations can be achieved through techniques such as BayesSHAP, BayesLIME, which extend traditional XAI methods, and CE, which provides both FI and confidence measures.

In particular, CE exhibits a significant limitation in multiclass contexts. It conventionally provides explanations only for whether an instance belongs to the predicted class or not (Löfström et al., 2024). As a result, this method fails to offer insight into the relationship of an instance with other possible classes, limiting its effectiveness in multiclass classification tasks.

Manufacturing machines are exposed to multiple types of failure, each requiring unique maintenance actions. The limitation of CE in explaining only the predicted class can be problematic in scenarios such as machine failure analysis, where understanding alternative outcomes is crucial, particularly when the predicted class has a probability of less than 0.5. For example, a model predicts "no failure" with a probability of 0.4, "failure type 1" with 0.35, and "failure type 2" with 0.25. CE would only explain the prediction for "no failure," despite the model having low confidence in that outcome. In such cases, it is more informative to provide explanations for "failure type 1" or even "failure type 2," especially if they carry higher risk or cost implications. To address this limitation, we extend CE to generate explanations not only for the predicted class, but also for other classes that are non-negligible or of critical interest.

This work contributes to uncertainty-aware explainable ML by extending CE to multiclass prediction settings:

- We extend CE from binary to multiclass classification using a one-vs-rest (OvR) strategy (Johansson et al., 2021), overcoming the limitation of explaining only the most probable class and enabling simultaneous interpretation of multiple competing failure modes.
- We propose a novel explanation design that displays (i) calibrated probability intervals for all classes in a compact upper panel, and (ii) class-specific FI bars with uncertainty shading for each factual condition. This allows users to assess both model confidence and explanatory uncertainty across multiple risk scenarios in a single glance.

- We perform a comprehensive quantitative evaluation of multiclass explanation methods, comparing extended CE with SHAP and LIME in terms of stability, robustness, and explanation generation time across multiple predictive maintenance datasets.
- We empirically assess how probabilistic predictions, uncertainty intervals (UIs), and explanation formats influence trust, DM behavior, and explanation preference in high-stakes industrial settings, providing the first human-grounded evaluation of uncertainty-aware multiclass explanations.

2. Background

A wide range of methods have been proposed to explain black-box ML models (Guidotti et al., 2018). Many of these approaches aim to approximate the behavior of complex models locally or globally using interpretable surrogates.

Among the most widely used techniques is LIME (Ribeiro et al., 2016), which explains individual predictions by fitting a local linear model around a target instance. Similarly, SHAP (Lundberg and Lee, 2017) provides feature attributions based on Shapley values, offering a theoretically grounded and consistent framework for local explanations. These methods have become standard tools due to their model-agnostic nature and ease of use.

Beyond linear explanations, alternative approaches have been proposed to improve interpretability. For example, Anchors (Ribeiro et al., 2018) generate rule-based explanations in the form of high-precision if-then statements, while MAPLE (Plumb et al., 2018) combines local linear models with ensemble methods to improve fidelity. Other approaches, such as AIME (Nakanishi, 2023) and RuleMatrix (Ming et al., 2019), focus on improving interpretability through inverse modeling or interactive rule-based visualizations.

More recent work has emphasized the importance of UQ in XAI, aiming to improve the reliability and trustworthiness of model explanations.

Bayesian extensions of popular methods, such as BayesLIME and BayesSHAP (Slack et al., 2021), provide probabilistic FI estimates by incorporating uncertainty into the explanation process. Similarly, BMB-LIME (Hung and Lee, 2024) enhances LIME by integrating nonlinear modeling and bootstrap aggregation within a Bayesian framework, improving both stability and fidelity.

Other approaches address uncertainty from different perspectives. For example, EATU (Huang et al., 2022) combines DL and game-theoretic explanations to improve stability in anomaly detection tasks, while (Löfström et al., 2024) introduces CE, which quantifies uncertainty in both FI and model predictions.

Despite these advances, uncertainty-aware explanation methods remain relatively underexplored compared to traditional deterministic approaches, highlighting the need for more robust and reliable XAI techniques.

3. Contribution - Extended Multiclass CE

CE can generate several types of explanations, including factual calibrated explanation (FCE) and alternative calibrated explanations (ACE).

To overcome an existing limitation in multiclass handling within CE, we extended both FCE and ACE using an OvR approach. We also redesigned the visualization to improve

interpretability and support better user trust, especially in DM contexts where users need to assess multiple failure risks simultaneously (see Section 5.1).

A VA calibrator (Vovk and Petej, 2012) was used by Löfström et al. (2025) to obtain the calibrated probability estimate of the positive class for CE binary classification, obtaining both lower and upper bounds for each estimate. The final numeric estimate was computed using the regularized mean of these bounds. In more formal terms, CE was used to produce explanations for binary classification as follows:

1. A pretrained scoring classifier h was used to produce a score s when predicting an object $h(x)$.
2. A VA was applied to generate a calibrated prediction p and an uncertainty interval with a lower and upper bound $[p_{\text{low}}, p_{\text{high}}]$ for the positive class.
3. For each feature f :
 - (a) Systematically change the value of feature f to produce slightly perturbed versions of object x . Then apply the calibrator from step 2 to estimate the averaged prediction $p.f$ of UIs $[p_{\text{low}}.f, p_{\text{high}}.f]$ for the perturbed objects.
 - (b) FI for feature f was defined as the difference between the calibrated prediction p , achieved on the original object x , and the averaged calibrated prediction $p.f$.
 - (c) UIs for FI were similarly defined by calculating the difference between p and UIs $[p_{\text{low}}.f, p_{\text{high}}.f]$ for the perturbed versions of x .
 - (d) A factual condition was formed for each feature.

To extend CE to multiclass problems, Löfström et al. (2024) applied the OvR method in step 2 of CE binary classification. For each class label c in the set of all classes C , a VA was initialized by treating class c as the positive class and using the probability estimate for class c as the score s^c from the base model. The output was the predicted class label $c = \arg \max_{c \in C} p^c$, along with the probability interval $[p_{\text{low}}^c, p_{\text{high}}^c]$ and the regularized probability p^c . Additionally, the output included the VA, defined using g_c and $g_{\neg c}$, which is used in step 3 of the CE rule generation process.

Löfström et al. (2024) applied step 3 only to the predicted class label, while our approach was to apply step 3 to each class label $c \in C$, generating an explanation containing feature weights for each class label c . The generated explanations of the object x were sorted based on the factual conditions and labeled by their class labels for visualization purposes.

For the combined multiclass point estimates to be usable as a probability distribution, it is sufficient to require $\sum_{c \in C} p^c = 1$. This is achieved by scale-normalizing all bounds and point estimates after calibration, preserving the per-class uncertainty signal in the interval widths.

VA (Vovk and Petej, 2012) extends naturally to multiclass classification via an OvR decomposition. For a problem with class set C , a separate binary VA calibrator is trained for each class $c \in C$, using a binarized target ($y = c$).

Each calibrator produces a validity interval $[p_{\text{low}}^c, p_{\text{high}}^c]$ and a point estimate $p^c = \phi(p_{\text{low}}^c, p_{\text{high}}^c)$, where ϕ is an interval summarization function. Because the binary OvR calibrators operate independently, the raw intervals carry no joint constraint across classes.

A key property of the raw VA intervals is that interval widths $D^c = p_{\text{high}}^c - p_{\text{low}}^c$ vary naturally across classes: classes for which the calibrator is more uncertain produce wider intervals. We preserve this per-class difficulty signal by applying a uniform *scale normalization*: all bounds and point estimates are divided by the same scalar $S = \sum_{k \in C} p^k$, the row sum of the raw point estimates:

$$p_{\text{low}}^c \leftarrow \frac{p_{\text{low}}^c}{S}, \quad p_{\text{high}}^c \leftarrow \frac{p_{\text{high}}^c}{S}, \quad p^c \leftarrow \frac{p^c}{S}, \quad \forall c \in C.$$

This enforces $\sum_c p^c = 1$ exactly while preserving the relative ordering of interval widths across classes. Because all quantities are divided by the same factor, the displayed bounds remain consistent with the normalized point estimates.

Scale normalization ensures that the point estimates sum to one and preserves the relative ordering of interval widths. However, because the normalization factor is data dependent, the normalized bounds are not claimed to retain the original validity properties of the binary Venn–Abers intervals. We therefore interpret them as normalized uncertainty summaries.

4. Experimental Setup

4.1. Experimental Process

This study followed a structured experimental pipeline consisting of four phases.

- Datasets and Models Selection
- CE extension implementation
- Quantitative evaluation experiments
- User survey for explanation assessment

4.2. Datasets and Models Selection

In this research, four predictive maintenance datasets were used to evaluate the performance of the explanation methods:

- Predictive Maintenance of Machines (CIA) ([H, 2024](#)): This dataset provides detailed information about engine performance and potential indicators of failure.
- Pump sensor data (Pump) ([UnknownClass, 2019](#)): This dataset includes timestamps, machine status, and readings from 52 sensors.
- Microsoft Azure Predictive Maintenance (Azure) ([Arnab, 2021](#)): This dataset contains data on machine conditions and usage from sensors, historical failure data, records of past maintenance (repairs and replacements), and machine-specific characteristics such as engine size, make, model, and location.

- AI4I 2020 Predictive Maintenance (AI4I) (Matzka, 2020): This dataset comprises 10,000 rows, each with 14 features. The data covers various aspects of machine operation, including a unique identifier, product ID, type, air temperature, process temperature, rotational speed, torque, tool wear, a binary indicator of machine failure, and a machine failure label indicating the occurrence of one of five failure modes: tool wear failure (TWF), heat dissipation failure (HDF), power failure (PWF), overstrain failure (OSF), or random failure (RNF).

Four predictive models (classifiers) were also selected to compare the performance of the explanation methods:

- DT (Decision Tree): A simple and interpretable model suitable for XAI, previously noted for its strong performance in machine failure prediction (Matzka, 2020; Sinha et al., 2023; Torcianti and Matzka, 2021; Alvanpour et al., 2020).
- GRU (Gated Recurrent Unit): GRU was chosen for its ability to handle time series data and effectively manage the gradient explosion problem common in traditional recurrent neural networks (Xu et al., 2022; Lin et al., 2023).
- RF (Random Forest): This model uses an ensemble of DTs to produce more robust and accurate predictions. It effectively handles high-dimensional data and helps reduce the risk of overfitting (Alam et al., 2024).
- XGB (XGBoost): An advanced ML technique known for its exceptional predictive power and versatility across various applications (Alam et al., 2024).

4.3. Quantitative Evaluation Experiments

This evaluation compared our extended FCE and ACE methods with two widely used post-hoc XAI techniques:

- SHAP: Used to provide feature-level insight into model predictions by estimating the contribution of each feature.
- LIME: Applied to generate locally faithful explanations for individual predictions.

All experiments were conducted using the four datasets described in Subsection 4.2. For each dataset:

- 20 instances were randomly selected for explanation testing, stratified by class distribution.
- The remaining data was divided into 67% for training and 33% for calibration and explanation.
- Each experiment was repeated 30 times for statistical robustness.

The four explanation approaches, extended FCE, extended ACE, SHAP, and LIME, were assessed. Each was evaluated on three key criteria:

- Stability: Consistency of explanations across similar inputs.

- Robustness: Resistance to minor input perturbations.
- Explanation Generation Time: Efficiency in producing explanations.

Extended FCE and ACE were applied to classifiers calibrated using VA calibration, reflecting the integrated uncertainty-aware nature of the framework. SHAP and LIME were tested on both uncalibrated and VA-calibrated versions of the GRU, DT, RF, and XGB models.

In these experiments, LimeTabularExplainer v0.2.0.1, SHAP v0.46.0, and CE v0.11.0 were used.

All experiments were carried out using a personal laptop with an Intel(R) Core(TM) i7-6700HQ CPU @ 2.60GHz, 16.0 GB of RAM and 64-bit Windows 10 Home.

4.4. Evaluation Metrics

To assess the quality and practicality of the framework’s explanation capabilities, we conducted a thorough evaluation focusing on three key aspects: stability, robustness, and explanation time. These criteria were chosen to ensure that the generated explanations are not only informative but also reliable and efficient for real-world applications.

Stability refers to how consistent the explanations’ outputs are for the same input data in multiple runs (Slack et al., 2021). Ideally, similar inputs should lead to similar explanations. Stability is a fundamental requirement for any explanation technique (Löfström et al., 2024). It was measured by the variance between the most important features for prediction in multiple runs. Low variance means high stability.

Robustness evaluates how well the explanations hold up even when the input instance or its surrounding context is slightly altered (Doshi-Velez and Kim, 2017). The aim is to ensure that explanations remain meaningful and do not significantly degrade under challenging conditions. In this study, robustness is operationalized narrowly as sensitivity of the most influential feature weight across repeated runs with varying random seeds; it should not be interpreted as a complete measure of explanation robustness.

Explanation Time measures the time it takes to generate explanations for a given input during the prediction process. This is crucial for time-sensitive applications where interpretability is needed in real time.

Stability and *robustness* results were reported as the average variance of FI weights for the most influential feature, measured over 30 runs and averaged over 20 test instances. The variance was calculated for each instance for the feature that was most frequently assigned the highest absolute importance. This feature is most likely to influence decisions and is expected to show the greatest variation due to its prominence in the explanations.

4.5. User Survey for Explanation Assessment

This survey investigated how the inclusion of prediction probabilities, UIs, and multiple multiclass explanation techniques in model output affected user trust, DM, and explanation preferences.

A within-subjects design was employed, meaning that each participant interacted with all explanation formats. This enabled a direct comparison of trust, DM, and explanation

preferences within the same individuals across all explanation types. This approach was advantageous because it:

- Reduced variability caused by individual differences in trust perception ([Charness et al., 2012](#)).
- Required fewer participants to achieve sufficient statistical power ([Charness et al., 2012](#)), which was important given the limited availability of individuals in the target group (data scientists with machine failure prediction expertise, data engineers with manufacturing experience, and mechanical engineers).
- Enabled direct comparisons between participants across conditions.

Each participant was presented with four prediction scenarios representing different levels of failure risk (86%, 63%, 56% and 35%) for a specific type of failure, using six different explanation formats:

1. Binary classification without additional explanation (A failure is going to occur).
2. Multiclass classification without probabilities or further explanation (A specific type of failure will occur).
3. Multiclass classification showing probabilities for four failure types, without additional explanations or UIs.
4. Multiclass classification showing probabilities and FI weights for each failure type, accompanied by a visualized Kernel SHAP explanation (without UIs).
5. Multiclass classification showing probabilities and FI weights for each failure type, accompanied by a visualized LIME explanation (without UIs).
6. Multiclass classification showing probability ranges (UIs) and FI weights for each failure type, accompanied by a visualized extended FCE explanation.

For each scenario, participants were asked to:

- Decide whether to act on the model’s prediction (Stop production / Continue production), measuring DM.
- Rate their trust in each explanation format on a scale from 1 to 10, measuring perceived trust.
- Indicate whether they would revise any previous decisions after reviewing all explanation formats, capturing awareness and reflection on explanation techniques.
- Rank the explanation formats from most to least preferred.
- Indicate whether their decision would change if the cost of stopping production differed significantly from the cost of missing a failure.

The analysis focused on comparing preference rankings among SHAP, LIME, and FCE explanations, as well as examining differences in trust and DM across explanation formats. In particular, comparisons were made between multiclass classification without probabilities and with probabilities to evaluate the impact of probability information, and between SHAP and LIME explanations versus FCE explanations to assess the effect of incorporating UIs.

5. Results

5.1. Demonstration of Extended FCE

The original multiclass FCE visualization introduced by [Löfström et al. \(2024\)](#), used only two colors: red and blue. Red indicated a positive class prediction, and red FI bars showed a factual condition positively influencing the probability of the predicted class. Conversely, blue indicated a negative class prediction and blue FI bars showed a negative influence on the probability of the predicted class.

The new FCE visualization comprises two main sections. The top panel displays calibrated multiclass probabilities for a given instance. Each colored bar represents the calibrated probability of the corresponding failure class (indicated by the legend). The lighter shaded portion of each probability bar visualizes the calibrated uncertainty interval associated with that class prediction. This interval reflects the inherent uncertainty in the probability estimate after calibration.

The lower panels provide local FI explanations conditioned on the most relevant factual conditions for the predicted outcome. Each row corresponds to a specific factual condition. Within each row, FI for each class is represented by a colored bar (matching the color of the class label). The intensity of the color indicates the magnitude of the feature’s impact, while a lighter shade of the same color shows the uncertainty interval associated with that FI. All FI bars originate from a central zero line. A bar extending to the right signifies a positive influence of the factual condition on the probability of its corresponding class. Conversely, a bar extending to the left indicates a negative influence on that class probability. The actual value of the feature for the instance being explained is shown on the right, and the factual condition to which FI applies is displayed on the left. The uncertainty interval for FI provides an estimated range for the lower and upper bounds of each feature’s contribution to the prediction of each class.

Looking at the top box in [Figure 1](#), there are several colored bars: The red bar represents the PWF class and is the longest and extends to around 0.38 on the probability scale. This indicates that the model, after calibration, predicts a PWF with a probability of 38%, which is the regularized mean. The red bar also has a lighter red section extending from about 0.35 to 0.43. This is the calibrated uncertainty interval for the PWF prediction. It tells us that while the model’s regularized mean estimate is 38%, the normalized uncertainty summary ranges from approximately 35% to 43%; no formal coverage guarantee is claimed for this normalized range.

The other colored bars, light blue for the RNF class and yellow for the none class, are shorter, indicating lower calibrated probabilities for those failure types for this specific machine instance. They also have their own lighter shaded UIs.



Figure 1: Example of a Factual Calibrated Explanation for a multiclass failure prediction instance

Looking at one of the lower boxes in Figure 1, the one related to **Rotational speed** > -0.76 with an actual value of 6.16, there are colored bars originating from the central zero line within this box.

The red bar (for PWF) extends significantly to the right, to around 0.4. This means that when the rotational speed is greater than -0.76 (and in this case, it is quite high at 6.16), it has a strong positive impact on the probability of a PWF. The lighter red section around this bar shows the uncertainty in this positive influence, which is around 0. The lighter red section is very short and might not be seen in the rotational speed box, it can be seen easily in other boxes.

The light blue bar (for RNF) extends significantly to the right, to around 0.3. This means that when the rotational speed is greater than -0.76 , it has a positive impact on the probability of RNF. The yellow bar (for none) in this same "Rotational speed" box extends clearly to the left, approximately -0.74 . This would indicate that the same condition (high rotational speed) has a strong negative impact on the probability of none class.

The other very short colored bars in this "Rotational speed" box suggest that this specific condition of rotational speed does not significantly influence the probability of those other failure types for this instance.

5.2. Quantitative Evaluation Results

The stability and robustness of the extended FCE and ACE methods were compared with calibrated SHAP (SC), uncalibrated SHAP (SU), calibrated LIME (LC), and uncalibrated LIME (LU). Tables 1 and 2 present the average variance of the most influential feature

weights as a measure of stability and robustness, respectively. Table 3 shows the average explanation time per instance.

Table 1: Explainability Stability Comparison

Dataset	Algorithm	FCE	ACE	LC	SC	LU	SU
CIA	XGB	4e-33	4e-34	5e-05	2e-32	7e-07	3e-37
	RF	7e-33	9e-34	2e-05	4e-35	1e-06	1e-35
	DT	1e-35	1e-35	4e-08	2e-35	3e-05	3e-34
	GRU	3e-33	3e-33	2e-03	1e-31	2e-07	2e-35
Azure	XGB	1e-33	2e-34	1e-03	8e-31	1e-05	6e-15
	RF	3e-33	3e-34	6e-04	3e-31	1e-05	9e-10
	DT	9e-34	4e-34	6e-04	3e-31	5e-05	4e-09
	GRU	6e-33	1e-33	2e-03	8e-31	2e-05	8e-09
AI4I	XGB	2e-33	2e-34	6e-04	3e-31	3e-06	2e-35
	RF	4e-33	5e-34	5e-04	2e-31	3e-06	1e-35
	DT	8e-34	3e-35	5e-04	2e-31	2e-05	2e-34
	GRU	9e-33	2e-33	1e-03	3e-31	2e-06	2e-35
Pump	XGB	2e-33	1e-34	9e-04	6e-31	8e-06	6e-05
	RF	3e-33	8e-34	4e-04	2e-31	8e-06	6e-07
	DT	4e-33	3e-33	4e-04	2e-31	4e-05	4e-05
	GRU	6e-33	1e-33	1e-03	6e-31	5e-05	6e-05

The results in Table 1 indicate that both extended FCE and extended ACE exhibited the lowest average variance, suggesting better stability in their explanations compared to LC, LU and SC across all datasets, while they achieved close stability to SU. A t-test reveals that there are no statistically significant differences, in terms of stability, between extended FCE, extended ACE, and SU on larger datasets such as Azure and Pump. However, on smaller datasets like CIA and AI4I, extended FCE and extended ACE demonstrate significantly better stability than SU.

Table 2: Explainability Robustness Comparison

Dataset	Algorithm	FCE	ACE	LC	SC	LU	SU
CIA	XGB	0.0079	0.0068	0.0233	0.0119	0.0002	0.0001
	RF	0.0052	0.0037	0.0144	0.0087	0.0001	0.0000
	DT	0.0087	0.0077	0.0109	0.0066	0.0039	0.0037
	GRU	0.0054	0.0054	0.0142	0.0088	0.0002	0.0001
Azure	XGB	0.0028	0.0025	0.1493	0.0767	0.0001	0.0001
	RF	0.0021	0.0016	0.1105	0.0263	0.0001	0.0000
	DT	0.0030	0.0027	0.1093	0.0256	0.0025	0.0013
	GRU	0.0028	0.0027	0.1170	0.0545	0.0001	0.0000
AI4I	XGB	0.0042	0.0038	0.2220	0.0603	0.0001	0.0001
	RF	0.0030	0.0022	0.1653	0.0259	0.0001	0.0000
	DT	0.0045	0.0041	0.1636	0.0249	0.0020	0.0019
	GRU	0.0030	0.0029	0.1737	0.0271	0.0001	0.0000
Pump	XGB	0.0090	0.0038	0.1122	0.0576	0.0001	0.0001
	RF	0.0070	0.0032	0.0831	0.0198	0.0001	0.0000
	DT	0.0446	0.0162	0.0822	0.0193	0.0019	0.0012
	GRU	0.0043	0.0035	0.0880	0.0410	0.0002	0.0001

Regarding robustness in Table 2, extended FCE and extended ACE outperform LC and SC. However, LU and SU show greater robustness, reflected in lower variance, with SU achieving the highest level of robustness overall.

Table 3: Explanation Time Comparison

Dataset	Algorithm	FCE	ACE	LC	SC	LU	SU
CIA	XGB	0.0467	0.0645	0.0308	0.0108	0.0802	0.0575
	RF	0.0281	0.0441	0.0295	0.0066	0.0638	0.2037
	DT	0.0224	0.0368	0.0301	0.0064	0.0388	0.0080
	GRU	0.0302	0.0407	0.0322	0.0082	0.0522	0.0663
Azure	XGB	0.5534	0.8094	0.0510	0.0155	0.1256	0.0687
	RF	0.5363	0.8090	0.0495	0.0111	0.1037	0.2028
	DT	0.5166	0.7883	0.0531	0.0139	0.0743	0.0175
	GRU	0.5496	0.8507	0.0505	0.0092	0.0831	0.0465
AI4I	XGB	0.0537	0.0760	0.0280	0.0067	0.0857	0.0370
	RF	0.0322	0.0489	0.0264	0.0038	0.0706	0.1283
	DT	0.0245	0.0399	0.0267	0.0037	0.0389	0.0048
	GRU	0.0325	0.0487	0.0289	0.0048	0.0528	0.0382
Pump	XGB	0.5958	0.8871	0.0818	0.0177	0.1552	0.0685
	RF	0.5721	0.8741	0.0793	0.0134	0.1293	0.2073
	DT	0.5576	0.8594	0.0837	0.0156	0.1021	0.0199
	GRU	0.5861	0.9115	0.0812	0.0126	0.1166	0.0541

The results in Table 3 indicate that both extended FCE and extended ACE had the highest average explanation times, indicating lower performance in terms of computational efficiency. However, their explanation times were comparable to that of LU and SU on smaller datasets such as CIA and AI4I.

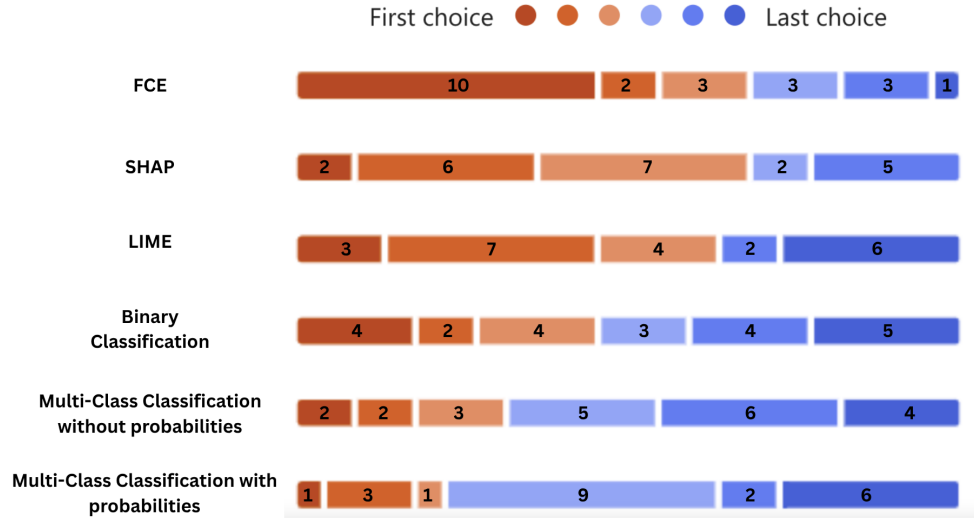


Figure 2: Participant Ranking of Model Result Presentation and Explanations

5.3. Survey Results

This section presents the findings from a survey examining how different explanation formats influence user trust, preferences, and DM in multiclass machine failure prediction. A within-subjects design was used, allowing for direct comparison across six types of explanations. A total of 22 domain experts completed the survey.

As shown in Figure 2, FCE was the most preferred explanation format, indicated by the longest red bar, followed by SHAP and LIME, respectively. Simpler formats, such as binary classification without explanation and multiclass classification without probabilities, were less preferred. Multiclass classification with probabilities was ranked the least preferred, indicated by the shortest red bar and the longest blue bar.

Participants rated their trust in each explanation format on a scale from 1 to 10. Table 4 shows the mean trust scores.

The lowest trust was observed for the multiclass classification without probabilities (Mean = 5.50) and the binary classification without explanations (Mean = 5.68). Adding probabilities increased trust (Mean = 6.55). Including FI further increased trust, with SHAP (Mean = 6.82) and LIME (Mean = 7.14) outperforming probability-only formats. Extended FCE achieved the highest average trust (Mean = 7.18), closely followed by LIME.

Table 4: Mean and Standard Deviation of Trust Ratings for Different Explanation Formats

	FCE	LIME	SHAP	Multiclass with probs	Binary	Multiclass without probs
Mean	7.18	7.14	6.82	6.55	5.68	5.50
STD	1.50	1.67	1.44	1.41	2.08	1.92

The participants decided whether to stop or continue production based on the model outputs in six explanation formats and four scenarios. Figure 3 summarizes the number of stop decisions per format.

FCE led to fewer stop decisions than the other formats, particularly against binary classification without probabilities, SHAP, LIME, and multiclass with probabilities. Multiclass with probabilities also resulted in fewer stop decisions than less detailed formats. LIME and SHAP showed intermediate behavior, while binary and multiclass without probabilities led to the highest number of stop decisions. Because the scenarios did not include observed failure outcomes, these differences indicate changes in decision thresholds rather than improvements in decision accuracy.

After reviewing all types of explanations, 45% of the participants reported that they would change at least one of their previous stop/continue-production decisions. This indicates that richer explanations can affect decision confidence. Participants often revised their choices based on perceived reliability and clarity, especially when uncertainty was included.

The thresholds for the decision of the participants were strongly influenced by the perceived cost of stopping production versus missing a failure. When the cost of stopping production was lower than the cost of missing a failure, participants were more willing to stop production even under uncertainty. Conversely, when stopping costs were high, participants required stronger evidence and higher confidence before deciding to stop production.

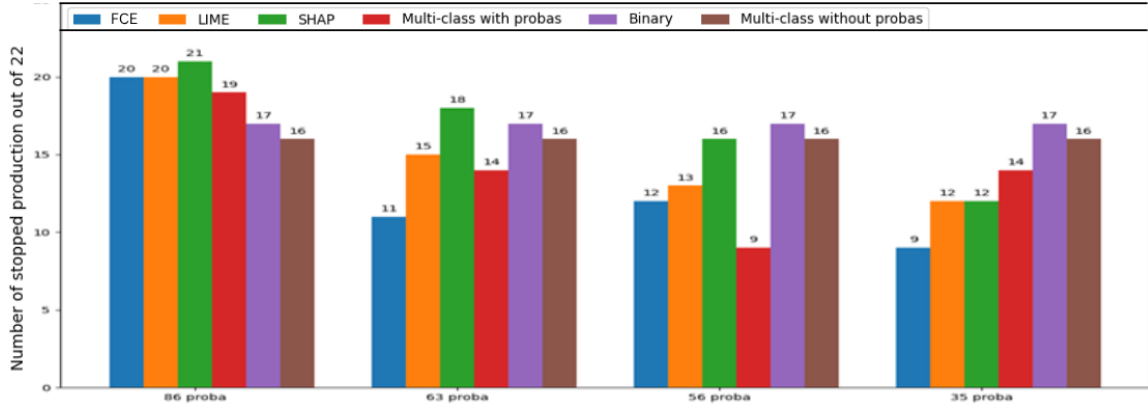


Figure 3: Number of Participants Choosing to Stop Production Across Explanation Formats

5.4. Qualitative Feedback

The open-ended responses provide qualitative insights that complement the survey’s quantitative findings. Several key themes emerge:

Preference for Visual Explanations (FCE, SHAP, LIME): Participants preferred extended FCE, SHAP, and LIME because their visualizations and explanations made decisions easier. Extended FCE was especially valued for giving a clear overview, allowing users to combine model outputs with their own experience, although some found UIs slightly confusing.

Value of UQ: The inclusion of UIs in extended FCE was widely appreciated, as it helped participants assess the reliability of predictions and increased their confidence in DM, even when failure probabilities were moderate. Several respondents explicitly highlighted uncertainty communication as a key factor in trusting the model, particularly in high-risk industrial settings.

Importance of FI: FI information, as provided by SHAP, LIME, and extended FCE, was considered highly valuable. It allowed participants to understand the reasoning behind the predictions and to evaluate whether the model’s behavior aligned with their domain knowledge. Explanations that were consistent with known physical or engineering relationships were perceived as more trustworthy. LIME was additionally valued for its instance-level interpretability.

Cost Sensitivity and Risk Assessment: Participants emphasized the importance of considering the costs associated with false positives (e.g., unnecessary production stops) and false negatives (e.g., missed failures). DM was often framed in terms of trade-offs between operational downtime and the risk of severe system failure. Some participants indicated a willingness to accept higher costs if explanations justified preventative actions.

Preference for Comprehensive Information: Overall, participants favored explanation formats that combined prediction probabilities, uncertainty information, and FI. This combination supported more informed and nuanced DM.

Cautious DM with Uncertainty: While uncertainty information improved understanding, it also led some participants to adopt a more cautious approach. In such cases, higher confidence and stronger supporting evidence were required before taking action.

Trust and Reliance on Explanations: Trust in the model was closely linked to the availability and clarity of explanations. Participants indicated that explanations increased their confidence and influenced their willingness to act on the model’s predictions, highlighting the importance of interpretability for practical adoption.

Desire for Improved Multiclass Visualization: One participant suggested that multiclass classification results could be presented more effectively, indicating that current visualizations may lack clarity in this setting.

Finally, it was observed that richer explanations, while beneficial, tended to increase DM time.

Taken together, the quantitative and qualitative findings suggest that uncertainty-aware multiclass explanations can improve both interpretability and user confidence, while also introducing additional cognitive demands. These observations motivate the broader conclusions and future directions discussed next.

6. Conclusions

This work addressed a critical limitation of CE, namely that it explains only the predicted class when explaining multiclass classifier predictions. By extending CE using an OvR strategy, we enable the generation of class-specific FI explanations and UIs for all competing failure modes. This extension provides practitioners with a more complete view of model behavior, particularly when the predicted class has low probability and alternative outcomes carry high risk.

Our empirical results show that this richer explanatory framework maintains strong reliability. Extended FCE and ACE deliver highly stable and robust explanations, outperforming LIME-based approaches and remaining competitive with SHAP. While this comes with increased computational cost, the trade-off is a substantially deeper and more informative view of model behavior.

The human-centered evaluation further highlights an important distinction: trust does not necessarily imply preference. Although extended FCE was the most trusted explanation format, not all users preferred it in practice. More detailed explanations, particularly those that incorporate uncertainty, tended to increase cognitive load and reaction time and were not universally favored. This reveals a fundamental tension in XAI: increasing informational richness can improve understanding and confidence, but may simultaneously reduce efficiency and usability for certain users or time-critical scenarios.

Despite its contributions, this work has several limitations. The computational overhead of extended FCE and ACE remains a barrier for real-time or resource-constrained deployments. While CE itself supports relatively efficient explanation generation, this efficiency has not yet been fully extended to the multiclass setting, where explanations are generated for all class labels sequentially. Additionally, the user study, while informative, involved a modest sample size and focused on manufacturing experts. Generalizability to other domains and user populations requires further investigation.

Future work should address these challenges along several directions. First, improving computational efficiency through smarter sampling, approximation methods, or incremental explanation updates is essential for real-time applicability. Second, adaptive explanation systems should be explored, where the level of detail, uncertainty representation, and visualization complexity are dynamically tailored to user expertise, task demands, and time constraints. Third, broader and more diverse user studies are needed to better understand how different stakeholders interact with uncertainty-aware explanations across domains. Finally, integrating interactive and user-driven explanation interfaces, where users can query, simplify, or expand explanations on demand, could help bridge the gap between trust, preference, and practical usability.

Beyond computational efficiency and usability considerations, another open challenge concerns the normalization of multiclass calibrated probability intervals. Different normalization strategies may affect both the probabilistic interpretation of the intervals and the preservation of uncertainty information. The following subsection discusses these alternatives and their associated trade-offs.

6.1. Normalization Strategies for Multiclass VA Intervals

This work used scale normalization (Section 3), which divides all bounds and point estimates by the same row sum to enforce $\sum_c p^c = 1$ while preserving the relative ordering of interval widths as a per-class uncertainty signal. However, several alternative normalization strategies exist, each making a different trade-off between probabilistic coherence of the displayed *bounds* and preservation of *interval width* as a difficulty indicator:

1. **Scale (used here):** Divide both bounds and point estimates by the common row sum $S = \sum_c p^c$. Point estimates sum to 1, the relative ordering of interval widths across classes is preserved, and bounds remain consistent with the normalized point estimates.
2. **Simplex:** Normalize only the point estimates to sum to 1; leave bounds at their raw VA values. Interval widths differ across classes and reflect genuine per-class calibrator uncertainty, but the displayed bounds are not rescaled to match the normalized point estimates.
3. **Coherence:** Enforce the additive constraint $p_{\text{high}}^c + \sum_{k \neq c} p_{\text{low}}^k = 1$ for every class c by adjusting the upper bounds: $p_{\text{high}}^c \leftarrow 1 - S_{\text{low}} + p_{\text{low}}^c$. This yields a stronger probabilistic statement about the bounds, but carries an important side effect: all class-wise interval widths become identical ($D = 1 - S_{\text{low}}$, where $S_{\text{low}} = \sum_c p_{\text{low}}^c$). This is not a design choice but a mathematical necessity: any exact additive coherence constraint of the form $h_c + \sum_{k \neq c} l_k = \alpha$ forces $D_c = \alpha - S_{\text{low}}$ for all c , independent of the individual l_c values. The per-class difficulty signal is therefore irrecoverably lost under this strategy.
4. **None (raw):** Return the raw VA outputs without modification. Useful for diagnostics; point estimates are not guaranteed to sum to 1.

The incompatibility between strategies 1–2 and strategy 3 is non-trivial: any scheme that enforces the exact additive coherence constraint must produce equal widths; conversely, any

scheme that preserves distinct widths cannot satisfy that constraint. Empirically evaluating how these strategies affect user trust and decision quality in explanation interfaces is a promising direction for future work.

Overall, this work moves XAI toward a more realistic paradigm: not just explaining predictions, but supporting human DM under uncertainty, while acknowledging that the “best” explanation is ultimately context dependent.

Acknowledgments

The authors acknowledge the Swedish Knowledge Foundation and industrial partners for financially supporting the research and education environment on Knowledge Intensive Product Realisation SPARK at Jönköping University, Sweden. Project PREMACOP grant no. 20220187.

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